DRAWING LIST FOR TEST PACKAGE

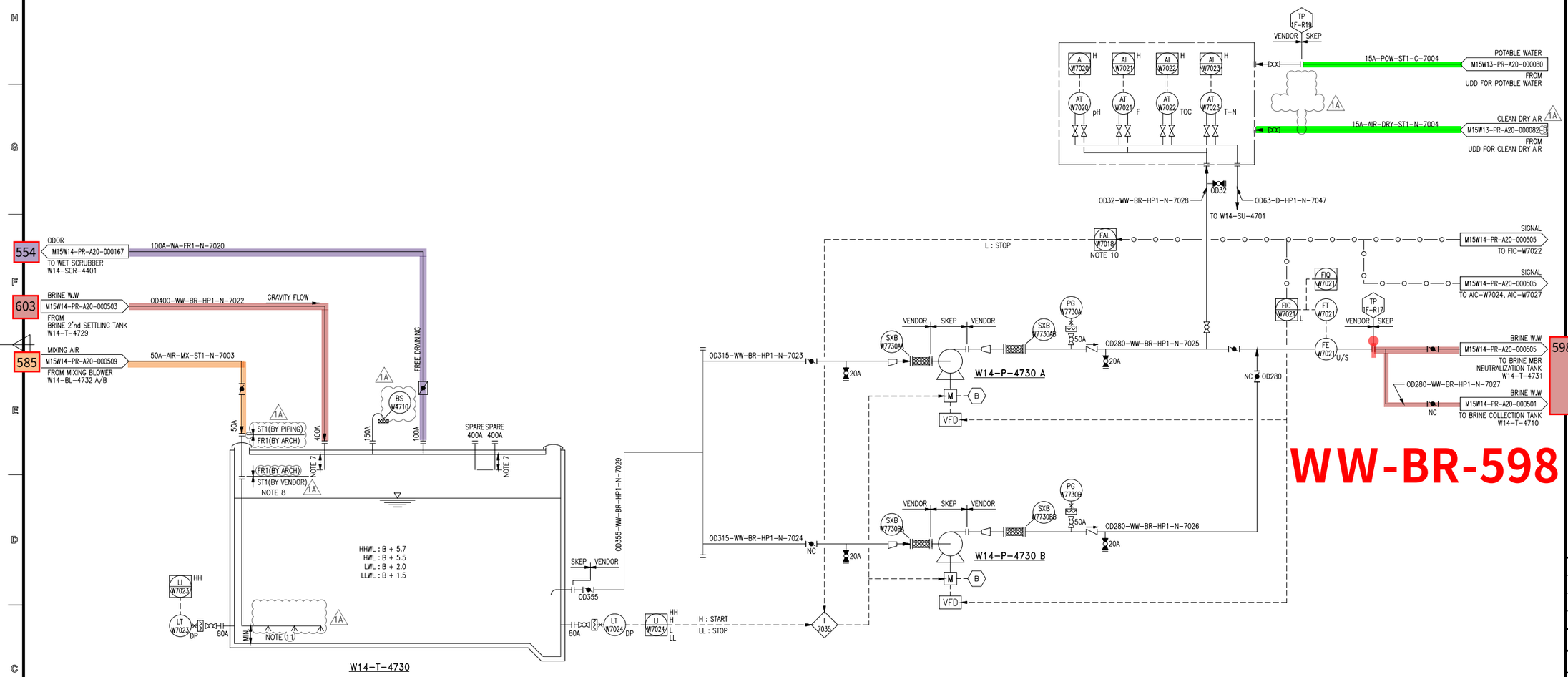
WW-BR-598

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	598	1FA	OD280-WW-BR-HP1-N-7025-02	2 OF 2	1	CJ-M15W14-PR-A20-000504, 505	1A/1A	
2	598	2FA	OD280-WW-BR-HP1-E-7025-01	1 OF 1	1	CJ-M15W14-PR-A20-000504, 505	1A/1A	
3	598	2FA	OD280-WW-BR-HP1-E-7027-01	1 OF 1	1	CJ-M15W14-PR-A20-000501, 504	1A/1A	

ITEM NO.		W14-T-4730
SERVICE		BRINE INTERMEDIATE TANK
Q'TY		1
TYPE		RECTANGULAR
SIZE	W x L x H | m	8.1 x 3.075 x 7.45 (EH 5.5)
CAPA.	m3	(185.6)(EH 137)
HRT	hr	0.7
MAT'L		P.C + FRP



ITEM NO.		W14-P-4730 A/B
SERVICE		BRINE INTERMEDIATE TRANSFER PUMP
Q'TY		1 + 1
TYPE		CENTRIFUGAL (VWVF)
CAPA.	m3/hr.ea x mH	250 x 30 (0.29MPa)
MAT'L		SSC13/SSC13
POWER	kW	(37)

대외비

KEY PLAN (1F)

NOTE

1. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것
2. 원수 인입지점은 펌프 흡입측에서 최대한 이격하여 설치
3. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것
4. MANHOLE : 2개 이상 설치한다. MANHOLE COVER : FRP
5. 온전시 CLOSE 되는 배관에는 LOTO 설치
6. CENTRIFUGAL PUMP MAT'L 표기 : CASING / IMPELLER
7. TANK 내 유입, 육안 확인이 가능하고, 비산되지 않도록 최소 깊이로 반영할 것 (SLAB 로 부터 100mm 띄어질 것)
8. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
9. INTERLOCK NO. I-7035 (W14-P-4730 A/B AUTO STOP) : REMOTE STATUS에서 W14-T-4730 LEVEL (LI-W7024 LOW LOW STOP 또는 W14-P-4730 A/B DISCHARGE FLOW (FAL-W7021) LOW STOP.
10. BRINE INTERMEDIATE TRANSFER PUMP RUN 후 30초 후 부터 동작한다.
11. DIFFUSER HOLE 및 배관배치의 관련된 상세 설계는 VENDOR가 수행한다.

REV.	DESCRIPTION	DATE
1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31

제 도 DRAWN BY	이 인 학
검 토 CHECKED BY	한 진 희
승 인 APPROVED BY	박 동 영
제 작 사	SK 에코플랜트
공 사 명 PROJECT TITLE	M15X Ph4 WWT Project
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM BRINE INTERMEDIATE TANK & TRANSFER PUMP
축 척 SCALE	NONE
날 짜 DATE	2025. 08. 14
PROJECT NO.	22330D
REV.	1A
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000504

C:₩Users₩TOP₩Desktop₩P&ID251013₩[M15X Ph-4 for 인허가]CAD FILE_0910₩CJ-M15W14-PR-A20-000504 R1 P&ID FOR BRINE

ITEM NO.		W14-T-4731
SERVICE		BRINE MBR NEUTRALIZATION TANK
Q'TY		1
TYPE		RECTANGULAR
SIZE	W x L x H | m	5.65 x 5.375 x (6.6 (EH 5.6))
CAPA.	m3	(200.4 (EH 170.1)) 1A
HRT	hr	(0.8) 1A
MAT'L		P.C + FRP

ITEM NO.		W14-T-4732
SERVICE		BRINE MBR 1'st ANOXIC TANK 1
Q'TY		1
TYPE		RECTANGULAR
SIZE	W x L x H | m	11.5 x 6.35 x (6.6 (EH 5.6))
CAPA.	m3	(482.0 (EH 408.9))
HRT	hr	(2.0) 1A
MAT'L		P.C + FRP

ITEM NO.		W14-T-4733
SERVICE		BRINE MBR 1'st ANOXIC TANK 2
Q'TY		1
TYPE		RECTANGULAR
SIZE	W x L x H | m	5.7 x 12.275 x (6.6 (EH 5.6))
CAPA.	m3	(461.8 (EH 391.8))
HRT	hr	(1.9) 1A
MAT'L		P.C + FRP

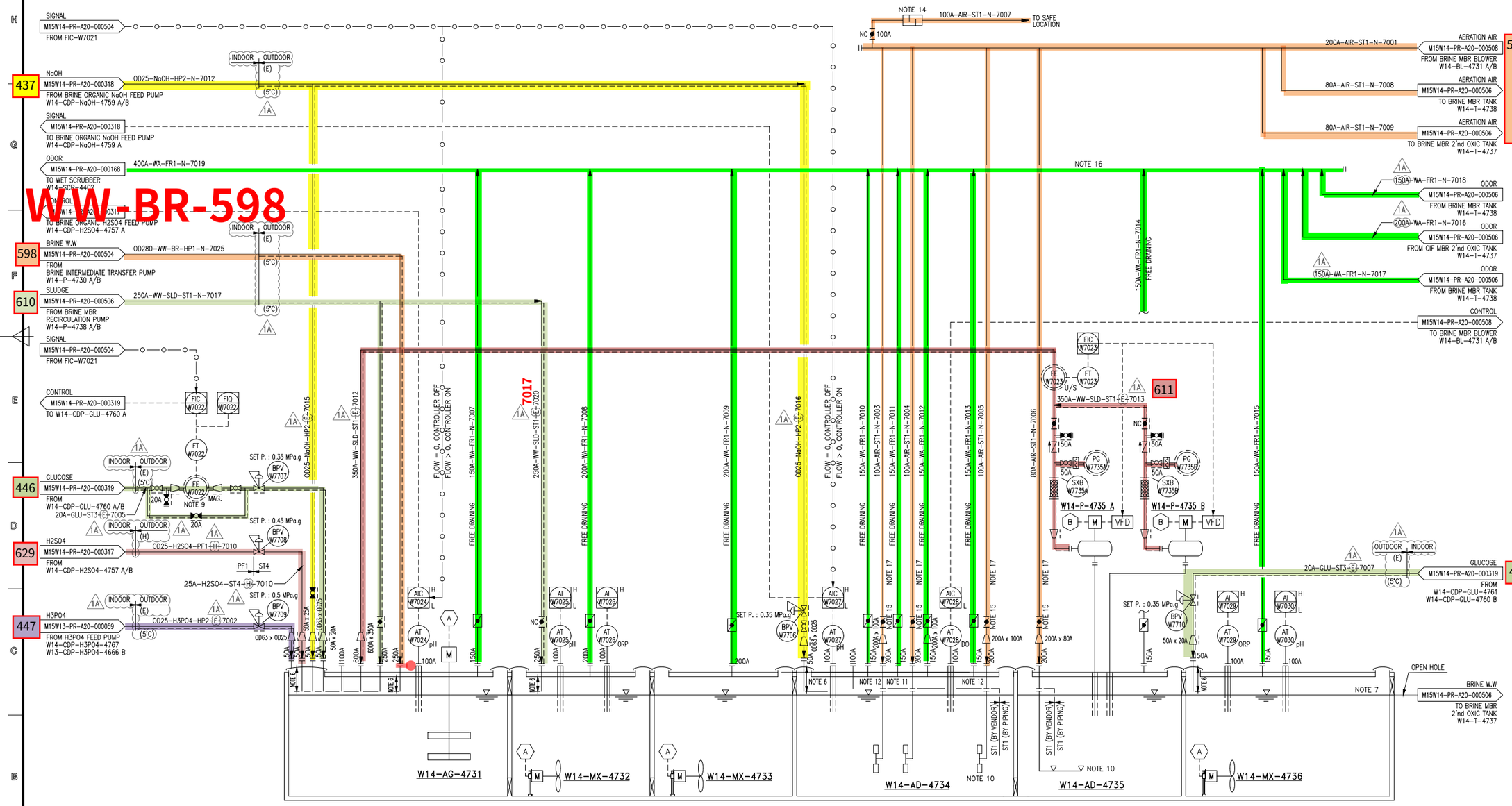
ITEM NO.		W14-T-4734
SERVICE		BRINE MBR 1'st OXIC TANK 1
Q'TY		1
TYPE		RECTANGULAR
SIZE	W x L x H | m	5.5 x 6.0 x (6.6 (EH 5.4))
CAPA.	m3	(217.8 (EH 178.2))
HRT	hr	(0.9) 1A
MAT'L		P.C + FRP

ITEM NO.		W14-T-4735
SERVICE		BRINE MBR 1'st OXIC TANK 2
Q'TY		1
TYPE		RECTANGULAR
SIZE	W x L x H | m	5.5 x 6.0 x (6.6 (EH 5.4))
CAPA.	m3	(217.8 (EH 178.2))
HRT	hr	(0.9) 1A
MAT'L		P.C + FRP

ITEM NO.		W14-T-4736
SERVICE		BRINE MBR 2'nd ANOXIC TANK
Q'TY		1
TYPE		RECTANGULAR
SIZE	W x L x H | m	5.625 x 6.0 x (6.6 (EH 5.2))
CAPA.	m3	(222.8 (EH 175.5)) 1A
HRT	hr	0.8
MAT'L		P.C + FRP

ITEM NO.		W14-AD-4734
SERVICE		BRINE MBR 1'st OXIC TANK AIR DIFFUSER 1
Q'TY	ea	(70) 1A
TYPE		MEMBRANE (TUBE)
CAPA.	L/min.ea	350
MAT'L		PP + EPDM

ITEM NO.		W14-AD-4735
SERVICE		BRINE MBR 1'st OXIC TANK AIR DIFFUSER 2
Q'TY	ea	(30) 1A
TYPE		MEMBRANE (DISC)
CAPA.	L/min.ea	100
MAT'L		PP + EPDM



W14-T-4731		W14-AG-4731
SERVICE		BRINE MBR NEUTRALIZATION AGITATOR
Q'TY		1
TYPE		HYDRO PADDLE NOTE 8
SPEED	rpm	1A (37)
MAT'L		STS316L/STS316L
POWER	kW	1A (11)

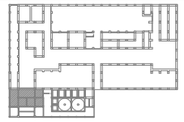
W14-T-4732		W14-MX-4732
SERVICE		BRINE MBR 1'st ANOXIC MIXER 1
Q'TY		1 1A
TYPE		SUBMERSIBLE
SPEED	rpm	(1800) 1A
MAT'L		STS431/STS304
POWER	kW	(11) 1A

W14-T-4733		W14-MX-4733
SERVICE		BRINE MBR 1'st ANOXIC MIXER 2
Q'TY		1 1A
TYPE		SUBMERSIBLE
SPEED	rpm	(1800) 1A
MAT'L		STS431/STS304
POWER	kW	(11) 1A

W14-T-4734		W14-P-4735 A/B
SERVICE		BRINE MBR SLUDGE INNER RECYCLE PUMP
Q'TY		1 + 1
TYPE		VERTICAL (VWF)
CAPA.	m3/hr.ea x mH	625 x 10 (0.10MPa)
MAT'L		SSC13/SSC13
POWER	kW	(37) 1A

W14-T-4736		W14-MX-4736
SERVICE		BRINE MBR 2'nd ANOXIC MIXER
Q'TY		1 1A
TYPE		SUBMERSIBLE
SPEED	rpm	(1800) 1A
MAT'L		STS431/STS304
POWER	kW	(5.5) 1A

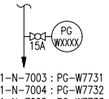
대외비



KEY PLAN (1MF)

NOTE

1. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
2. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
3. MANHOLE SIZE : 1000W x 1000L x 2EA/TANK MANHOLE COVER : FRP
4. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
5. AGITATOR MAT'L 표기 : SHAFT/BLADE
6. TANK 내 유입시, 육안 확인이 가능하고, 비산이 되지 않도록 쉬운 길이로 변형 할 것. (SLAB로부터 100mm 떨어질 것)
7. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
8. 추후 VENDOR 에서 결정한다.
9. INSTRUMENT VENDOR 선정 후 CONTROL SCHEME 및 CONTROL VALVE INLET / OUTLET NOZZLE SIZE 확정 예정.
10. CIF 1'st/2'nd OXIC TANK는 선회류 포기가 되도록 배열해야하나, 건물최적화에 따른 산기관 설치공간 부족으로 M15X PH4 WWT에는 적용 안함.
11. 1차 호기조 내 AIR 확인이 가능하도록 중간지점에 점검구 반영할 것.
12. 1차 호기조 내 pH, DO 계측기의 설치 위치는 중간 지점보다 앞쪽에 설치할 것.
13. CENTRIFUGAL PUMP MAT'L 표기 : CASING / IMPELLER
14. LOOSE SUPPLY INSTALLATION BY PURCHASER.
15. AIR량 조정을 위해 GEAR OPERATED BUTTERFLY VALVE를 적용한다.
16. 밸류기 배관 구성 및 배관 NUMBER는 도면번호 W14-PR-A20-000168을 기준으로 한다.
17. 유체의 흐름을 확인할 수 있도록 LINE 상에 PRESSURE GAUGE를 설치할 것. (TAG NO. 이래 참조)



200A-AIR-ST1-N-7003 : PG-W7731
200A-AIR-ST1-N-7004 : PG-W7732
200A-AIR-ST1-N-7005 : PG-W7733
200A-AIR-ST1-N-7006 : PG-W7734

REV.	수정 DESCRIPTION	날짜 DATE
1A	REVISED AFC	25.08.14
[illegible]	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31

제도 DRAWN BY	이 인 학
검토 CHECKED BY	한 진 희
승인 APPROVED BY	박 동 영
제작사	SK 에코플랜트
공사명 PROJECT TITLE	M15X Ph4 WWT Project
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM BRINE MBR ANOXIC & OXIC TANK
축척 SCALE	NONE
날짜 DATE	2025. 08. 14
PROJECT NO.	22330D
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000505
REV.	1A

WW-BR-598
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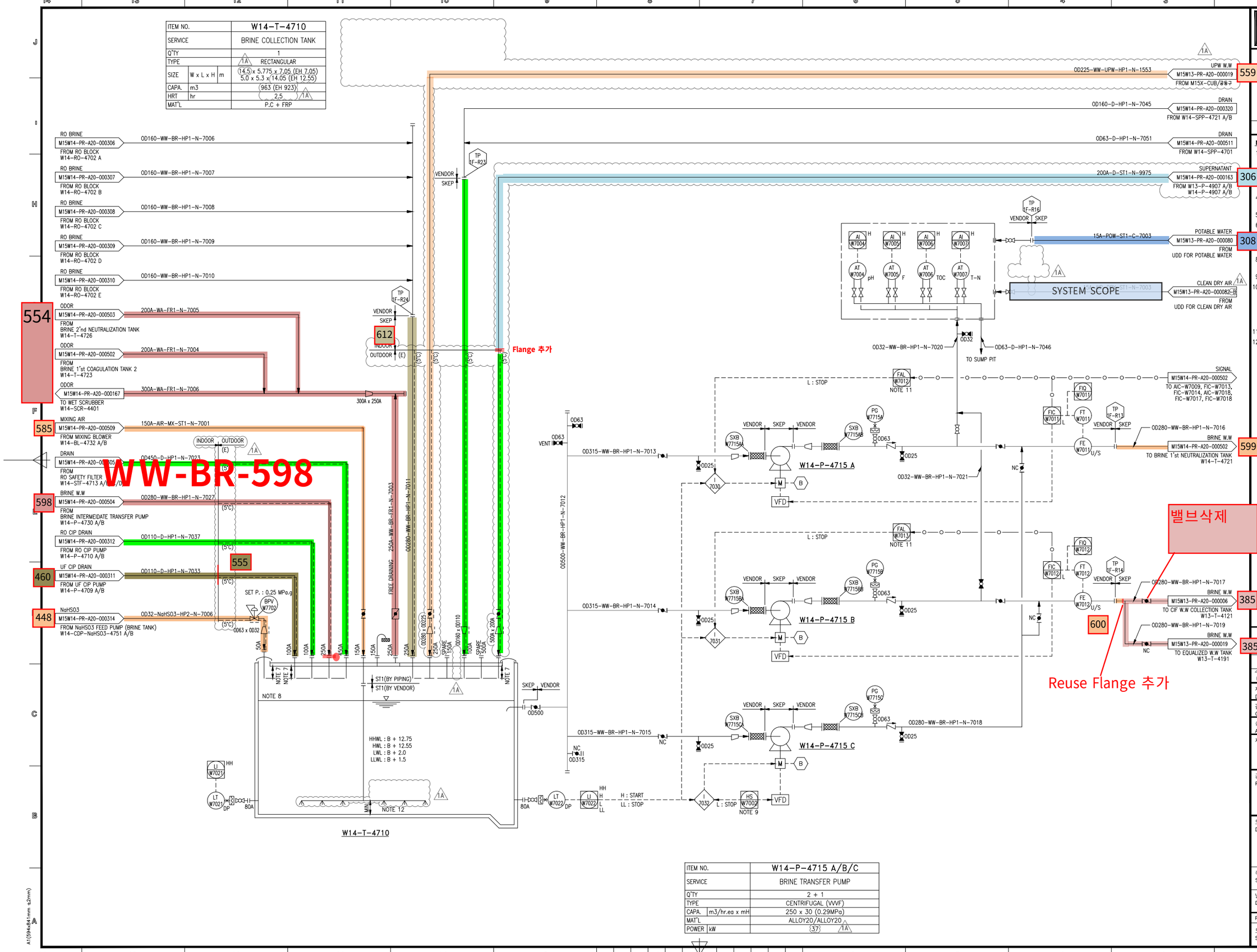
ITEM NO.		W14-T-4710
SERVICE		BRINE COLLECTION TANK
Q'TY		1
TYPE		RECTANGULAR
SIZE	W x L x H m	(14.5) x 5.775 x 7.05 (EH 7.05) 5.0 x 5.3 x (14.05 (EH 12.55)
CAPA.	m3	(963 (EH 923))
HRT	hr	2.5
MAT'L		P.C + FRP



ITEM NO.		W14-P-4715 A/B/C
SERVICE		BRINE TRANSFER PUMP
Q'TY		2 + 1
TYPE		CENTRIFUGAL (VVVF)
CAPA.	m3/hr.ea x mH	250 x 30 (0.29MPa)
MAT'L		ALLOY20/ALLOY20
POWER	kW	(37)

대외비

KEY PLAN (1F)

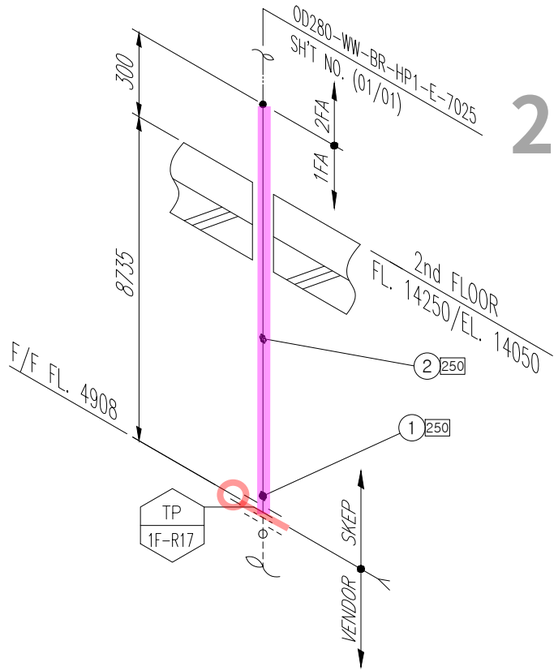
NOTE

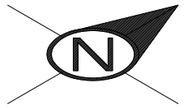
1. TANK와 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것
2. 원수 인입지점은 펌프 흡입측에서 최대한 이격하여 설치
   2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것
4. MANHOLE : 2개 이상 설치한다.
   MANHOLE COVER : FRP
5. 운전시 CLOSE 되는 배관에는 LOTO 설치
6. CENTRIFUGAL PUMP
   MAT'L 표기 : CASING / IMPELLER
7. TANK 내 유입, 육안 확인이 가능하고, 비산되지 않도록 최소 같이로 반영할 것 (SLAB 로 부터 100mm 탈이질 것)
8. 내부 FRP LINING TANK의 SLEEVE는 SUS304 + FRP LINING 적용.
9. FIC-W7011/W7012 중 선택운전.
10. INTERLOCK NO. I-7030/7031/7032 (W14-P-4715 A/B/C AUTO STOP)
    : REMOTE STATUS에서 W14-T-4710 LEVEL (LI-W7022) LOW LOW STOP 또는 W14-P-4715 A/B DISCHARGE FLOW (FAL-W7011/W7012) LOW STOP.
11. BRINE TRANSFER PUMP RUN 후 30초 후 부터 동작한다.
12. DIFFUSER HOLE 및 배관내치의 관련된 상세 설계는 (VENDOR)가 수행한다.

REV.	DESCRIPTION	DATE
1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.10.31

제 도 DRAWN BY	이 인 학
검 토 CHECKED BY	한 진 희
승 인 APPROVED BY	박 동 영
제 작 사	SK 에코플랜트
공 사 명 PROJECT TITLE	M15X Ph4 WWT Project
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM BRINE COLLECTION TANK & TRANSFER PUMP
축 척 SCALE	NONE
날 짜 DATE	2025. 08. 14
PROJECT NO.	22330D
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000501 REV. 1A

C:WUsersWTOPWDesktopWP&ID251013W[M15X Ph-4 for 인허가]CAD FILE_0910WCJ-M15W14-PR-A20-000501 R1 P&ID FOR BRINE





SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		HP1	
P	OD280	9.1M	PIPE, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
F	OD280	1EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
			LAP JOINT-STS304 / STUB END-HDPE PE100

<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT
SK hynix 1

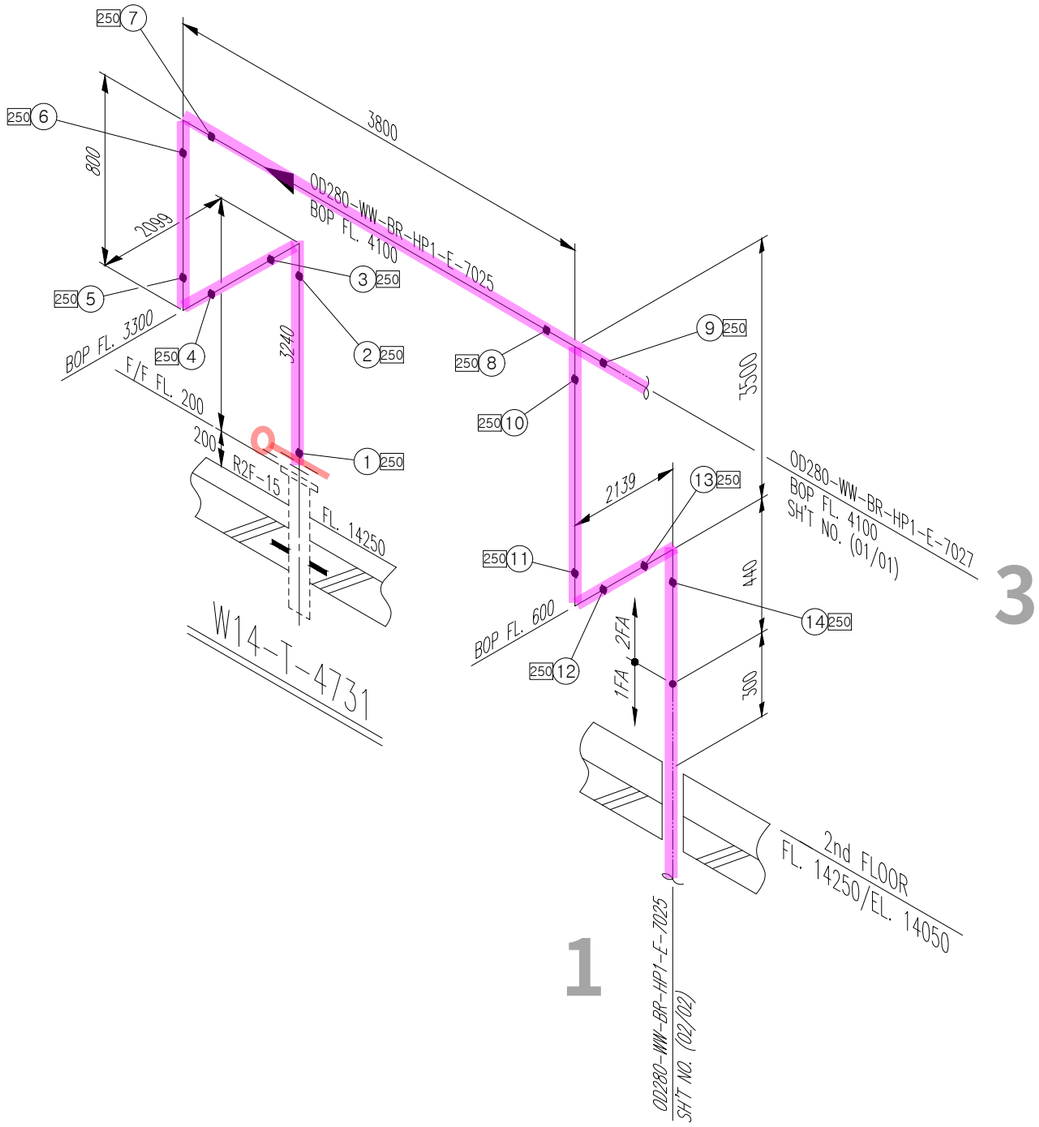
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	–	DESIGN	6.4	60
			OPERATING	4	AMB

AREA W14
UNIT R1FA
REF. PLAN DWG. CJ-M15W14-PI-A30-010017-B/C
REF. P & ID CJ-M15W14-PR-A20-000504

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
1	25.03.31	REVISED AS MARKED	황래종	안남후	박동영
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
PROJECT NAME M15X Ph4 Project
COMMODITY WW-BR
JOB NO. 22330D
LINE NO. OD280-WW-BR-HP1-N-7025
SH'T NO. 02 / 02
REV. 1

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



ITEM	SIZE	Q'TY	DESCRIPTION
		HP1	
P	OD280	16.0M	PIPE, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
E90	OD280	5EA	ELBOW 90 DEG, HDPE PE100, PE, BUTT FUSION,
			PN16/SDR11
T	OD280	1EA	TEE EQUAL, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
F	OD280	1EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
			LAP JOINT-STS304 / STUB END-HDPE PE100
G	OD280	1EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD280	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS
1 INC	25tXOD280	16.0M	관 보온재(고무발포) + 알루미늄 CASING

<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix 2

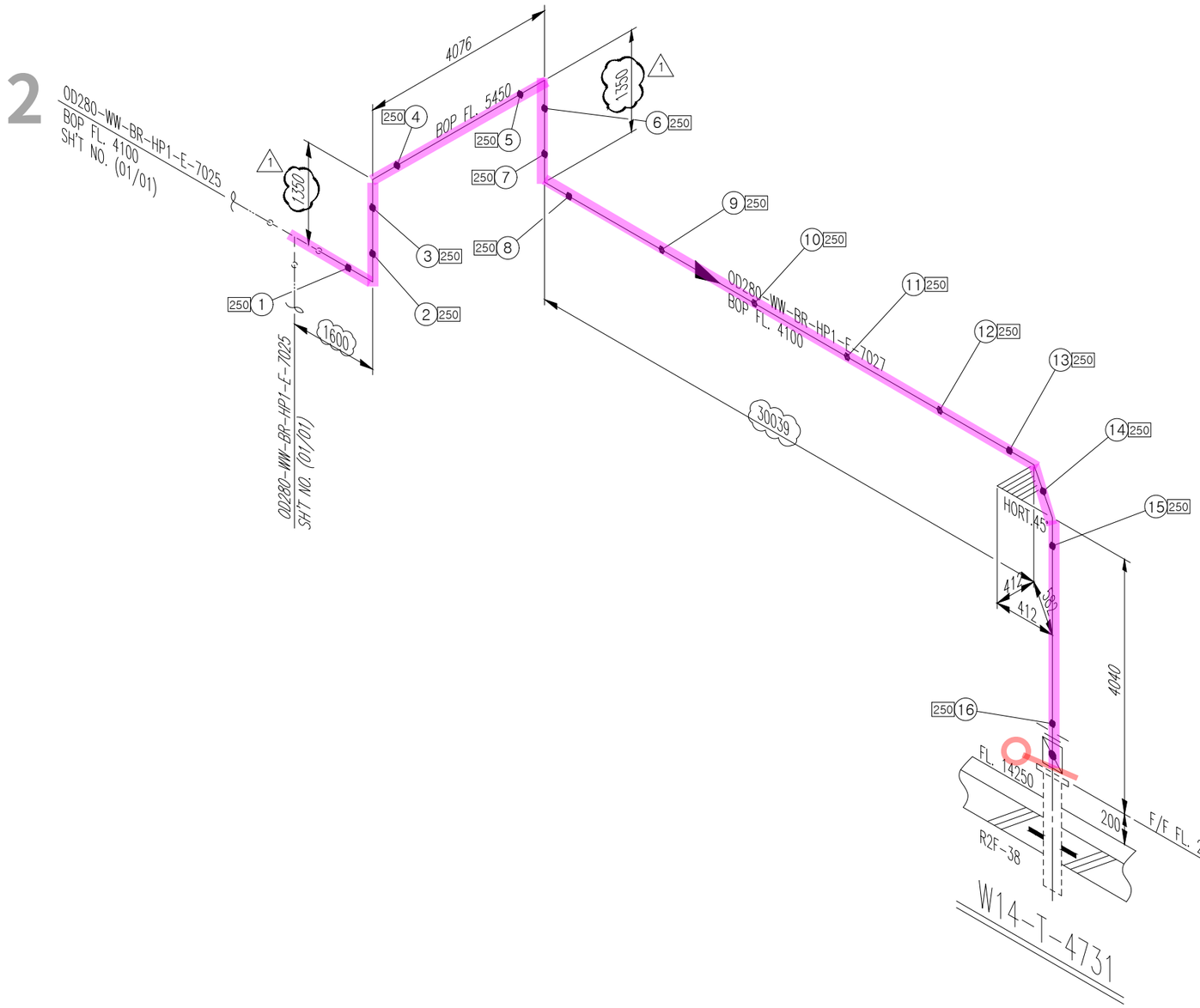
INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	25	DESIGN	6.4	60
			OPERATING	4	AMB

AREA W14
UNIT R2FA
REF. PLAN DWG. CJ-M15W14-PI-A30-020015-A
REF. P & ID CJ-M15W14-PR-A20-000504,000505

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
1	25.03.31	REVISED AS MARKED	황래종	안남후	박동영
0	24.12.20	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
PROJECT NAME M15X Ph4 Project
COMMODITY WW-BR
JOB NO. 22330D
LINE NO. OD280-WW-BR-HP1-E-7025
SH'T NO. 01 / 01
REV. 1

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		HP1	
P	OD280	1 41.9M	PIPE, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
E90	OD280	5EA	ELBOW 90 DEG, HDPE PE100, PE, BUTT FUSION,
			PN16/SDR11
E45	OD280	1EA	ELBOW 45 DEG, HDPE PE100, PE, BUTT FUSION,
			PN16/SDR11
VF	OD280	1EA	BUTTERFLY VALVE, KS 10K, WAFER FF, BODY-PP /
			TRIM-PP / EPDM-LINED, GEAR
F	OD280	1EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
			LAP JOINT-STS304 / STUB END-HDPE PE100
BL	OD280	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)
1 INC	25tXOD280	41.8M	관 보온재(고무발포) + 알루미늄 CASING 1

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBL0/VB1/VB1L0/VB2/VB2L0 * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDL0 * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL_PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT SK hynix 3

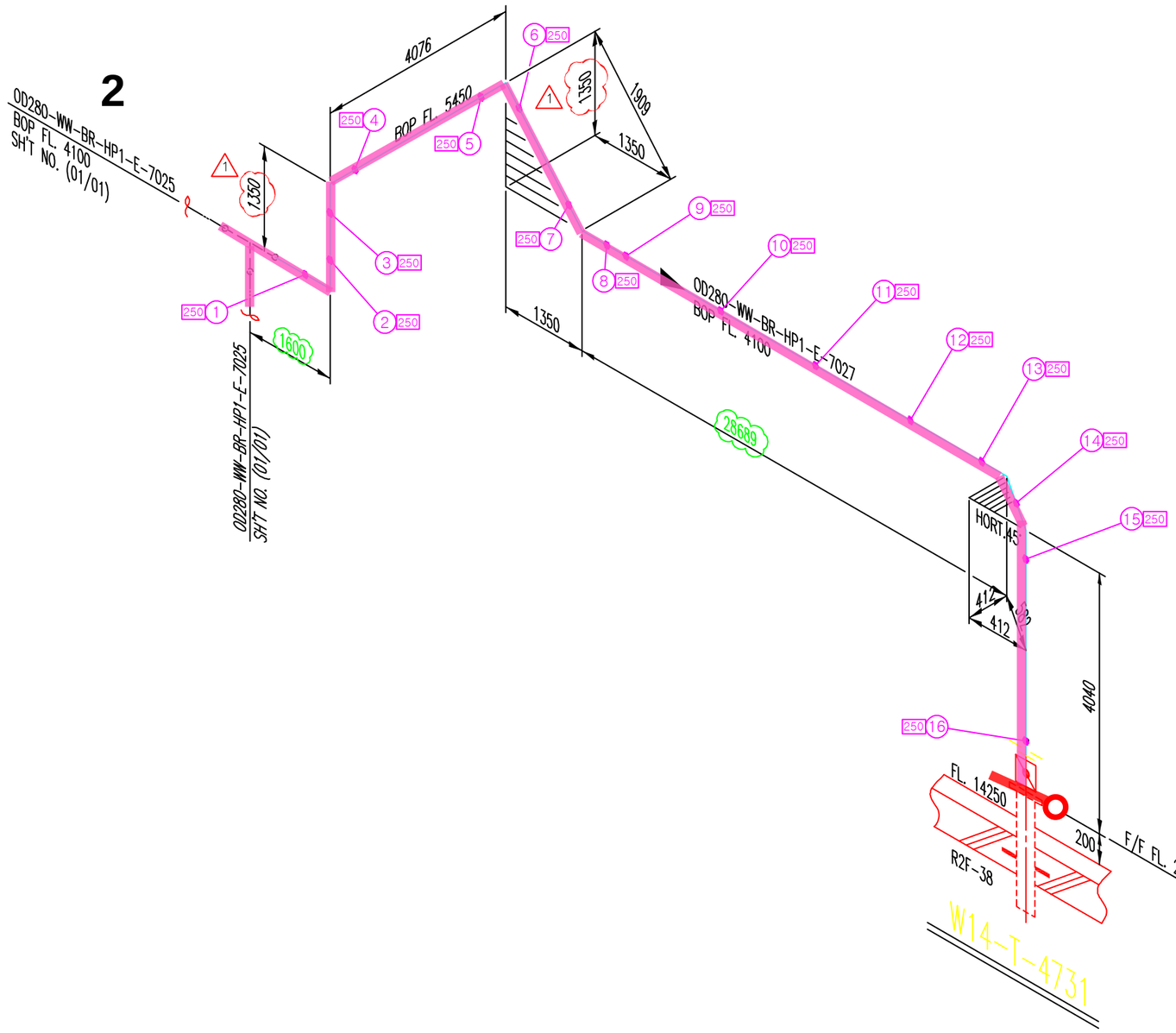
INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	25	DESIGN	6.4	60
			OPERATING	4	AMB

AREA W14
UNIT R2FA
REF. PLAN DWG. CJ-M15W14-PI-A30-020015-B
REF. P & ID CJ-M15W14-PR-A20-000501,000504

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
1	25.03.31	REVISED AS MARKED	황래종	안남후	박동영
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR SK ecoplant
PROJECT NAME M15X Ph4 Project
COMMODITY WW-BR
JOB NO. 22330D
LINE NO. OD280-WW-BR-HP1-E-7027
SH'T NO. 01 / 01
REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.



<NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

SK ecoplant

ITEM	SIZE	Q'TY	DESCRIPTION
		HP1	
P	OD280	1 41.9M	PIPE, HDPE PE100, PE, BUTT FUSION, PN16/SDR11
E90	OD280	4EA	ELBOW 90 DEG, HDPE PE100, PE, BUTT FUSION,
			PN16/SDR11
E45	OD280	2EA	ELBOW 45 DEG, HDPE PE100, PE, BUTT FUSION,
			PN16/SDR11
VF	OD280	1EA	BUTTERFLY VALVE, KS 10K, WAFER FF, BODY-PP /
			TRIM-PP / EPDM-LINED, GEAR
F	OD280	1EA	FLG LAP JOINT / STUB END SET, KS 10K, FF,
			LAP JOINT-STS304 / STUB END-HDPE PE100
BL	OD280	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)
1 INC	25tXOD280	41.8M	관 보온재(고무발포) + 알루미늄 CASING 1

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBL0/VB1/VB1L0/VB2/VB2L0 * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
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* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT: SK hynix

3

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)
	THICK.	25	DESIGN	6.4	60
			OPERATING	4	AMB

AREA: W14
UNIT: R2FA
REF. PLAN DWG.: CJ-M15W14-PI-A30-020015-B
REF. P & ID: CJ-M15W14-PR-A20-000501,000504

REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
1	25.03.31	REVISED AS MARKED	황래종	안남후	박동영
0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영

CONTRACTOR: SK ecoplant
PROJECT NAME: M15X Ph4 Project
COMMODITY: WW-BR
JOB NO.: 22330D
LINE NO.: OD280-WW-BR-HP1-E-7027
SH'T NO.: 01 / 01
REV.: 0